Gan Sang Wei, Wayne

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GitHub: https://github.com/xisingazix
Portfolio site: https://xisingazix.github.io

Career Summary

I bring over 10 years of transferable experience in analytical work, structured problem solving and process optimization, gain from my background in laboratory operations and military service. I am experienced in leading cross-functional teams, coordinating stakeholders and driving outcomes under time-sensitive conditions. I am detail-oriented, methodical and adaptable, with a strong focus on data integrity and workflow efficiency. I am now eager to apply these strengths to build reliable, user-focused web applications and contribute effectively as a developer in a growing tech environment.

Technical Skills

Client-side Languages: HTML5, CSS3, JavaScript (ES6+)
Server-Side Languages: Java, Python
Database: MySQL, SQL, Database Schema Design
Tools & Version Control: Git, GitHub, IntelliJ IDEA, Visual Studio Code, Postman
Testing & Debugging: Unit Testing (JUnit, PyTest), Debugging
Frameworks: React.js, Bootstrap, Spring Boot (Maven), Node.js.

Professional Development

Jr Full Stack Developer,
Generation Singapore (in partnership with **Microsoft & Temasek Poly)** | *2025–2026*

- Gained extensive hands-on experience across the entire development stack, Agile framework, MySQL, Norde.js and React.js

Capstone Project, link:
https://github.com/xisingazix/team3capstone.stayahead
https://github.com/xisingazix/capstone.stayahead.db

- Developed a full-stack financial education portal using Java Spring boot, JavaScript, and MySQL to promote scam awareness
- Engineered RESTful APIs with Spring Security for secure authentication and authorization, implementing data validation and JWT-based sessions
- Built a responsive front-end with HTML/CSS and interactive components featuring client-side form validation
- Designed database schema with proper entity mappings and constraints, tested endpoints using Postman
- Applied Version control and progress management using GitHub

Work Experiences

Senior Lab Analyst, Saybolt Singapore | *2023–2025*

- Applied systematic debugging and calibration practices to maintain high equipment reliability and ensure data integrity at approximately 99.5% accuracy.
- Managed data entry and quality assurance through a Laboratory Information Management System (LIMS), enforcing strict data validation and compliance protocols parallel to database management and secure coding practices.
- Utilized a methodical approach to test, troubleshoot and improve laboratory processes, demonstrating transferable problem-solving skills relevant to QA and code debugging

Lab Analyst, CCIC Singapore | *2021–2023*

- Optimized a complex hydrocarbon distillation process, applying data analysis to improve product yield by 4%, showcasing direct performance optimization and process improvement skills.
- Identified procedural gaps through documentation and root-cause reviews, reducing operational discrepancies and improving team guidance.
- Collaborated with lab teams to streamline workflows and increase throughput efficiency

Lab Analyst, Intertek Testing Services | *2020–2021*

- Applied analytical troubleshooting to maintain and calibrate complex laboratory systems
- Systematized large-scale data verification and documentation, ensuring high levels of accuracy and adherence to strict procedural protocols

Ammunition Specialist, Singapore Armed Forces | *2012–2020*

- Coordinated systems and resources across 86 facilities, strengthening structured problem-solving and dependency management.
- Led teams of up to 40 personnel and streamlined operations through data analysis, achieving a 25 percent cost reduction.
- Operated as Fire Safety Officer for Overseas facilities, conducted fire drills and training
- Achieved no major finding in entire overseas tour, rating of A for both years

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Education

Diploma in Chemical Engineering – Temasek Polytechnic | *2009–2012*
Specialization in Petroleum Processes
Trained and certified at Chemical Process Technology Centre (CPTC)